

Module 2 Assignment: Impact/Sustainability Questions

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Sustainability Questions:

Section 1: Organizational Values		
Questions	Answers	Current Industry Practices
What are your core values and the organizational culture you aspire to create?	SPARK's core values are intelligence, transparency, and transformation. We aim to build a culture of curiosity, responsibility, and systems-level thinking.	Many energy startups focus on speed-to-market and technical prowess but often lack intentional organizational cultures. B Corp and mission-driven companies like BlocPower, Opower, and Arcadia are known for integrating impact into culture from the start.
Who are your key stakeholders, and how do you plan to engage them?	Customers (building operators), utilities, regulators, investors, and tech partners. We engage through pilots, user-centered design, and data transparency.	Stakeholder engagement is often shallow in hardware/energy companies—limited to sales and compliance. Leading firms like SPAN and Nest have elevated stakeholder engagement by building intuitive interfaces and strong user communities.
What kind of impact beyond the bottom line do you want to make? (e.g. social, environmental)	Reducing waste, enabling intelligent infrastructure, and lowering energy burdens for customers.	Climate-tech firms increasingly include impact metrics (e.g., GHG avoided, job creation) in investor and customer materials. LACI portfolio companies like Homeboy Recycling integrate circularity and social equity.
Define the overarching purpose for DEIJBBA and selected focus.	To ensure the benefits of intelligent energy systems are inclusive and equitably distributed, with a focus on workforce and user access.	Many cleantech firms are early in DEIJBBA. Progressive orgs like GRID Alternatives and Elemental Excelsior lead the field in inclusive workforce development and community-based co-design.
Section 2: Public Policy Risks and Opportunities		
Questions	Answers	Current Industry Practices
What are existing local/state/federal/international regulations that have a DIRECT impact (positive or negative) on your license to operate?	Energy benchmarking ordinances, Title 24, building codes, and load management rules.	Many startups don't anticipate regulation early; others like ClearTrace and Voltus design explicitly for compliance. Title 24 has catalyzed innovation in smart HVAC and building controls.
What are existing regulations that have an INDIRECT impact on your license to operate?	Incentives for solar, DR, and smart building tech shape demand indirectly.	Smart thermostat companies (Nest, Ecobee) and DER aggregators actively track incentive structures to shape deployment.

Are there any pending new or changing regulations that could impact your business?	Yes — mandatory building performance standards, DR mandates, and local electrification targets.	Leading firms monitor rules through DSIRE, EIA, and utility filings. Regulatory risk is often underappreciated in early stages.
Which communities or regions are supportive of your idea? Which are not?	Denver, NYC, California cities — due to strong energy performance goals and funding.	Many firms pilot in California (CPUC support), NY (REV initiative), or Colorado (Xcel programs). Rural or fossil-fuel-dominant areas may resist load control tech.
Section 3: Social License to Operate		
Questions	Answers	Current Industry Practices
Who are your stakeholders DIRECTLY impacted by your business? How are they impacted?	Building tenants and operators: experience energy savings, automation, and cost visibility.	Few energy tech startups directly involve tenants. Programs like Elevate Energy and BlocPower prioritize tenant benefits and affordability.
Who are your stakeholders INDIRECTLY impacted by your business? How are they impacted?	Utilities, local communities, installation workforce.	Most tech firms prioritize customers only. Community impact is typically externalized unless part of a nonprofit or mission-aligned enterprise.
What alliances could you possibly form to improve positioning? e.g. Greenpeace, World Wildlife Fund, etc.	Greenlining, city green banks, community climate orgs, building coalitions, utility programs.	LACI members often partner with workforce nonprofits, city innovation offices, and NGOs for pilot access and credibility. GRID Alternatives is a model for public-private-alignment.
Section 4: Product/Service Design, Material/Chemical Use, and Public Health		
Questions	Answers	Current Industry Practices
What design framework are you using? (C2C, Biomimicry, DfE, LCA, Green Chemistry, other)	We incorporate Design for Environment (DfE) and modular design principles, with intent to conduct lifecycle assessment (LCA) once field data is available.	Most cleantech devices adopt modularity for servicing and repair. Companies like SPAN and Nest focus on intuitive, safe design; few conduct LCA at early stages.
What materials/chemicals are in your product/service, and what are their risks? e.g. toxicity, availability, intensity, price elasticity, recyclability, handling, volatility, etc.	The IEH uses common industrial electronics: circuit boards, sensors, enclosures, fasteners. Risks include electronics disposal, embodied carbon, and rare metal sourcing.	Hardware companies often rely on off-the-shelf components without traceability. More advanced companies (e.g., Fairphone) track supply chain ethics and toxicity.
How can you design with the end in mind? e.g. less/lighter material, less toxicity, recyclable & recycled, etc.	SPARK's product is designed for field upgradability, low-toxicity housing, and minimal permanent integration to reduce retrofit waste.	"Design for disassembly" is gaining traction in consumer electronics and building systems. Few startups plan for full product lifecycle due to cost.
Section 5: Clean Production: Energy, Land, Air, and Water Use		
Questions	Answers	Current Industry Practices
What is your largest source of energy use?	Device manufacturing and server/cloud analytics (once scaled).	Energy use is rarely accounted for in early-stage IoT. Data centers and embedded energy in supply chains are often untracked.

What is your largest source of land use?	Minimal — IEH mounts on an existing electrical panel.	Smart infrastructure devices typically reuse wall or panel space; land use is mostly tied to manufacturing partners.
Do you have air pollutants? If yes, what are they?	No direct emissions from device use. Potential scope 3 emissions from suppliers.	Most early-stage hardware firms do not track this; air pollution is more relevant in high-volume industrial production.
Where in your process can you practice clean production? (e.g. use renewable energy, recycled materials, etc.)	Using low-energy fabrication methods, selecting suppliers who use recycled materials, and hosting cloud services powered by renewable energy.	Increasingly common among companies working with partners like AWS Clean Energy or regional manufacturers using green processes.
Section 6: Supplier Networks		
Questions	Answers	Current Industry Practices
What are your UPSTREAM suppliers? (Existing or anticipated)	Electronics manufacturers (circuit boards, sensors, power supply), enclosure fabricators.	Most startups outsource to contract manufacturers in the U.S. or Asia. Environmental transparency varies.
What are your DOWNSTREAM suppliers? (Existing or anticipated)	Potentially channel partners for installation, utility program operators, service contractors.	Clean tech firms often rely on HVAC or solar installers for last-mile deployment. Few vet downstream impacts.
Do any of your suppliers consider environmental or human impacts in their operations? If yes, how so?	We intend to screen suppliers for compliance with environmental and labor standards and prioritize those with certifications (e.g., ISO 14001, B Corp).	Leading companies include supplier codes of conduct. Tesla, HP, and Apple publish audits, but startups rarely have leverage or resources early on.
Develop a Supplier Diversity Plan.	SPARK will prioritize working with minority-owned suppliers and contractors, and seek out diverse-owned businesses in component sourcing, installation, and operations.	Most early-stage cleantech firms do not formalize diversity plans. LACI encourages this in its ecosystem.
Becoming Minority Certified (if relevant, for your own company).	We are evaluating certification with the National Minority Supplier Development Council (NMSDC) or WBENC if applicable.	Certification is rare for hardware/software startups, though some pursue it for access to government or utility contracts.
Section 7: Post Use: Reuse, Repurpose, Return to Raw Materials		
Questions	Answers	Current Industry Practices
Categorize your waste stream by percentage: Hazardous, Recyclable, Reusable, Landfill.	Initial estimates: 5% hazardous (electronics), 65% recyclable (metal, plastic), 20% reusable (housing), 10% landfill.	Few early-stage firms track waste categorization. Leading examples include HP's take-back programs and Isidore's e-waste model.
What is the primary component of your waste stream?	Electronics (circuit boards and sensors), plastic enclosures.	E-waste is a growing concern in cleantech and consumer tech. Modular design and take-back systems are rare at early stage.
Are there any less harmful material substitutes you could use?	Yes — using recyclable polymers and minimizing mixed-material packaging can reduce downstream impact.	Most hardware startups prioritize cost over material sustainability unless incentivized by partners or customers.

What are ways to create a source of new materials through a take-back program?	Partner with certified e-waste recyclers to offer a free return option at end-of-life, or create a refurbishment/upcycling path through local orgs.	Take-back programs are emerging in consumer electronics and building hardware (e.g., Nest thermostat recycling) but uncommon among B2B startups.
LACI Just Impact Report		
Question	Answer	
Which metrics described in the Just Impact report would be applicable to your business?	- Jobs Created (PI3687): from installation, servicing, and internal roles. - GHG Reduction (PI5376): energy waste avoided through intelligent dispatch. - Non-Hazardous Waste Avoided (PI8177): through reducing overbuilt infrastructure. - Revenue Generated (PI1775): reflects viability of the business model. - Minority/Previously Excluded Employment (OI3236): via workforce development partnerships.	
What other metrics have you identified (if any) that would show greater impact, but which are not in the Just Impact report?	- Energy Optimization per Square Foot: reflects direct building efficiency impact. - Coordination Hours Enabled: a measure of energy decision-making intelligence (new metric). - Installation Time Saved: due to plug-and-play design. - Customer Control Events Logged: quantifies user empowerment in energy management.	